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#include <stdio.h>

int matA[20][20], matB[20][20], result[20][20];
int rA = 0, cA = 0, rB = 0, cB = 0;
int resultRow = 0, resultCol = 0;

void acceptMatrix() {
    printf("Enter rows and columns for Matrix A: ");
    scanf("%d %d", &rA, &cA);
    printf("Enter elements of Matrix A:\n");
    for (int i = 0; i < rA; i++) {
        for (int j = 0; j < cA; j++) {
            scanf("%d", &matA[i][j]);}}

    printf("Enter rows and columns for Matrix B: ");
    scanf("%d %d", &rB, &cB);
    printf("Enter elements of Matrix B:\n");
    for (int i = 0; i < rB; i++) {
        for (int j = 0; j < cB; j++) {
            scanf("%d", &matB[i][j]);}}}

void display() {
    printf("Result Matrix:\n");
    for (int i = 0; i < resultRow; i++) {
        for (int j = 0; j < resultCol; j++) {
            printf("%d\t", result[i][j]);}

        printf("\n");}}

void addMatrix() {
    if (rA != rB || cA != cB) {
        printf("Addition not possible. Dimensions must match.\n");
        resultRow = 0; resultCol = 0;
        return;}

    resultRow = rA;
    resultCol = cA;
    for (int i = 0; i < rA; i++) {
        for (int j = 0; j < cA; j++) {
            result[i][j] = matA[i][j] + matB[i][j];}}
    printf("Matrices added.\n");}

void subMatrix() {
    if (rA != rB || cA != cB) {
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        printf("Matrices have invalid dimentions.\n");
        resultRow = 0; resultCol = 0;
        return;}

    resultRow = rA;
    resultCol = cA;
    for (int i = 0; i < rA; i++) {
        for (int j = 0; j < cA; j++) {
            result[i][j] = matA[i][j] - matB[i][j];}}

    printf("Matrices subtracted.\n");}

void mulMatrix() {
    if (cA != rB) {
        printf("Matrices have invalid dimentions.\n");
        resultRow = 0; resultCol = 0;
        return;}

    resultRow = rA;
    resultCol = cB;
    for (int i = 0; i < rA; i++) {
        for (int j = 0; j < cB; j++) {
            result[i][j] = 0;
            for (int k = 0; k < cA; k++) {
                result[i][j] += matA[i][k] * matB[k][j];}}}

    printf("Matrices multiplied.\n");}

void transMatrix() {
    int choice;
    printf("Transpose which matrix? (1 for Matrix A, 2 for Matrix B): ");
    scanf("%d", &choice);

    if (choice == 1) {
        resultRow = cA;
        resultCol = rA;
        for (int i = 0; i < rA; i++) {
            for (int j = 0; j < cA; j++) {
                result[j][i] = matA[i][j];}}

        printf("Matrix A transposed.\n");}
    else if (choice == 2) {
        resultRow = cB;
        resultCol = rB;
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        for (int i = 0; i < rB; i++) {
            for (int j = 0; j < cB; j++) {
                result[j][i] = matB[i][j];}}
        printf("Matrix B transposed.\n");}
    else {
        printf("Invalid choice.\n");
        resultRow = 0; resultCol = 0;}}

int main(){
int choice;
do{
        printf("\nEnter operation:\n");
        printf("1) Enter Matrix.\n");
        printf("2) Display Result.\n");
        printf("3) Add Matrices.\n");
        printf("4) Subtract Matrices.\n");
        printf("5) Multiply Matrices.\n");
        printf("6) Transpose Matrix.\n");
        printf("7) Exit.\n");
        printf("Choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                acceptMatrix();
                break;
            case 2:
                display();
                break;
            case 3:
                addMatrix();
                break;
            case 4:
                subMatrix();
                break;
            case 5:
                mulMatrix();
                break;
            case 6:
                transMatrix();
                break;
            case 7:
                break;
            default:
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            printf("Invalid\n");}}
            while (choice != 7);

    return 0;}
```

# OUTPUT:

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Enter operation:
1) Enter Matrix.
2) Display Result.
3) Add Matrices.
4) Subtract Matrices.
5) Multiply Matrices.
6) Transpose Matrix.
7) Exit.
Choice: 1
Enter rows and columns for Matrix A: 2 2
Enter elements of Matrix A:
1 2 3 4
Enter rows and columns for Matrix B: 2 2
Enter elements of Matrix B:
4 5 6 7

Enter operation:
1) Enter Matrix.
2) Display Result.
3) Add Matrices.
4) Subtract Matrices.
5) Multiply Matrices.
6) Transpose Matrix.
7) Exit.
Choice: 3
```